

Right-Triangle Route Planner



Grade 8 • Mathematics
NC.8.G.7

Use the Pythagorean theorem and its converse.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.



Build a second model together

A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Discuss: What should the first step find?



Your turn • 1

A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Represent → solve → explain → check



Your turn • 2

A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Represent → solve → explain → check



Your turn • 3

A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Represent → solve → explain → check



Your turn • 4

A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Represent → solve → explain → check



Exit question • Explain independently

A route goes 9 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 39 m

2. 75 m

3. 75 m

4. 26 m

Exit: 15 m

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

